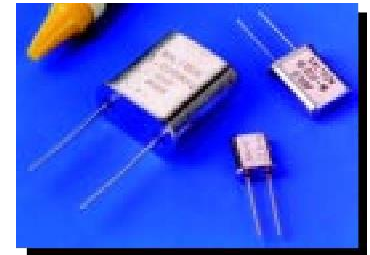


● FEATURES

- EXCELLENT CLOCK SIGNAL GENERATOR FOR CPU's
- WIDE FREQUENCY RANGE
- INDUSTRY STANDARD PACKAGES

● SPECIFICATIONS

FREQUENCY RANGE		100.00 kHz TO 300.00 MHz
HOLDER TYPES	A: HC-49/U	1.00 MHz TO 300.00 MHz
	B: HC-51/U	100.00 kHz TO 4.00 MHz
	C: UM-1	8.00 MHz TO 300.00 MHz
FREQUENCY TOLERANCE AT 25 °C		±30 PPM TYPICAL ±50 PPM MAXIMUM
FREQUENCY STABILITY OVER TEMPERATURE		±50 PPM TYPICAL ±100 PPM MAXIMUM
OPERATING TEMPERATURE RANGE		-20 °C TO +70 °C STANDARD
		-40° C TO +85 °C EXTENDED
STORAGE TEMPERATURE RANGE		-55 °C TO +125 °C
AGING		±5 PPM PER YEAR MAXIMUM
LOAD CAPACITANCE		10 TO 32 pF 0R SERIES
SHUNT CAPACITANCE		7 pF MAXIMUM
EQUIVALENT SERIES RESISTANCE (ESR)		SEE PAGE 11
DRIVE LEVEL		0.1 mW TYPICAL 1 mW MAXIMUM
SHOCK RESISTANCE		±5 PPM MAXIMUM 75 cm DROP TEST IN 3 AXES ONTO A HARD SURFACE OR 1000 g's x 0.5 ms x ½ SINEWAVE IN 3 AXES



● FREQUENCY RANGE AND MODE OF OSCILLATION BY HOLDER TYPE

HOLDER TYPE	FUNDAMENTAL (FUND)	THIRD OVERTONE (3OT)	FIFTH OVERTONE (5OT)	SEVENTH OVERTONE (7OT)	NINTH OVERTONE (9OT)
TYPE A: HC-49/U	1.00 MHz TO 65.00 MHz	20.00 MHz TO 90.00 MHz	60.00 MHz TO 130.00 MHz	125.00 MHz TO 200.00 MHz	180.00 MHz TO 300.00 MHz
TYPE B: HC-51/U	100.00 kHz TO 4.00 MHz				
TYPE C: UM-1	8.00 MHz TO 65.00 MHz	25.00 MHz TO 90.00 MHz	60.00 MHz TO 130.00 MHz	125.00 MHz TO 200.00 MHz	180.00 MHz TO 300.00 MHz

● PART NUMBERING SYSTEM

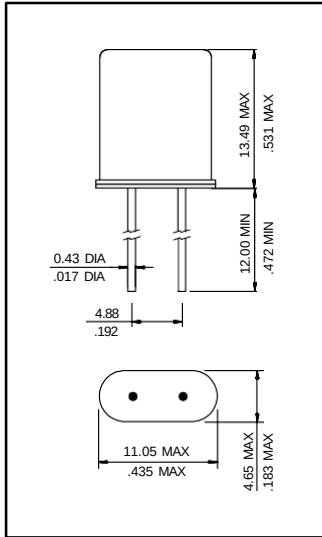
HOLDER TYPE		FREQUENCY	LOAD CAPACITANCE	MODE	EXTENDED TEMPERATURE
A B C	HC-49/U HC-51/U UM-1				
		IN MHz	10 TO 32pF FOR PARALLEL S FOR SERIES	F 3OT 5OT 7OT 9OT FUNDAMENTAL THIRD OVERTONE FIFTH OVERTONE SEVENTH OVERTONE NINTH OVERTONE	BLANK FOR STANDARD EXT FOR EXTENDED
MECHANICAL OPTIONS				TAPE AND REEL	
THREE PIN BASE		THIRD LEAD (TOP OF HOUSING)	SPACER	SLEEVE	TR
3PIN		3RD	SP	SL	

† FUNDAMENTAL MODE NOT NEEDED IN PART NUMBER

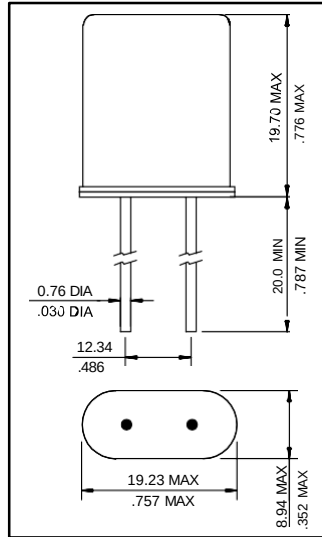
EXAMPLE: A-4.000-S

A-3.579545-18-EXT-3PIN-SP-SL-TR

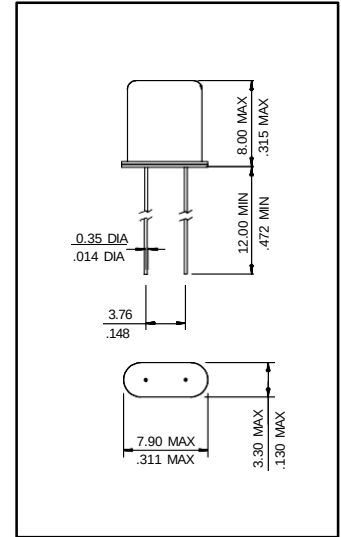
C-65.000-S-30T



TYPE A (HC-49/U)

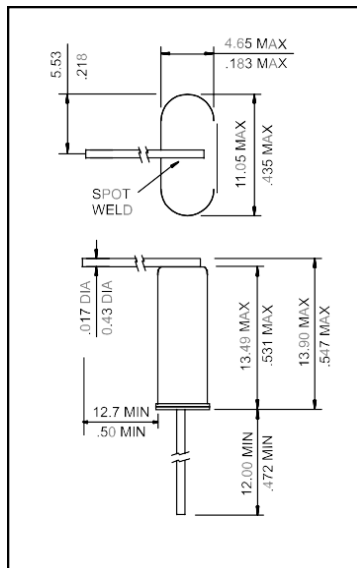


TYPE B (HC-51/U)

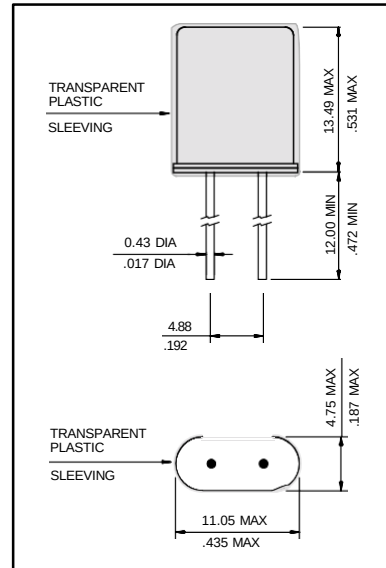


TYPE C (UM-1)

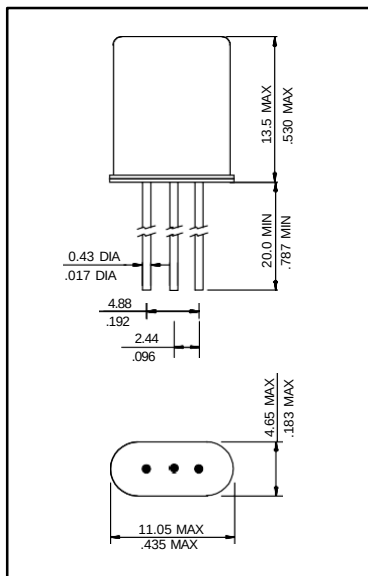
● PACKAGE OPTIONS



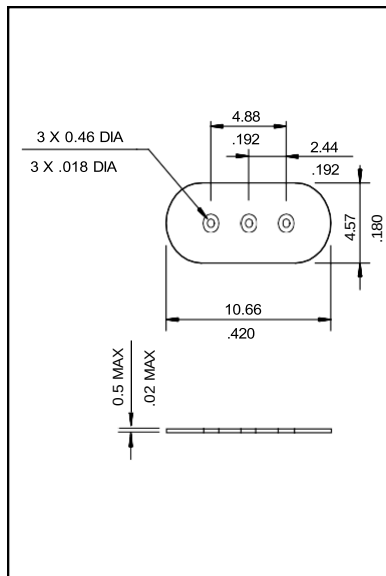
THIRDALEADOPTION: 3RD



PLASTICSLEEVEOPTION: SL



3LEADBASE OPTION: 3PIN



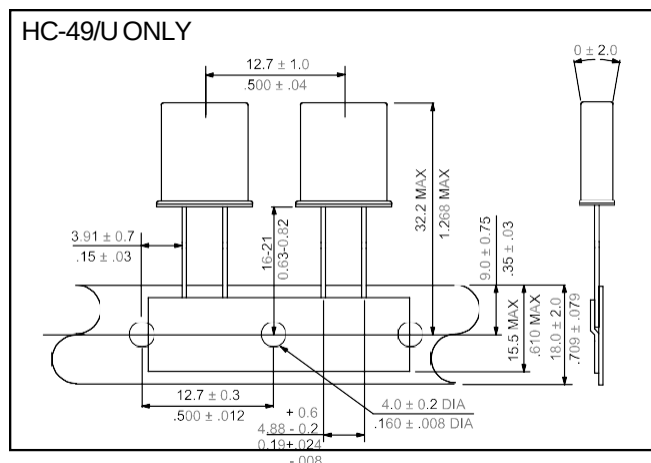
PLASTICSPACEROPTION: SP

● EQUIVALENT SERIES RESISTANCE VERSUS FREQUENCY AND PACKAGE

FREQUENCY MHz	HOLDER TYPE			ESR			
	HC-51/U "B"	HC-49/U "A"	UM-1 "C"	"B"	"A"	"C"	
1.000	B	A		700	700		
1.843	B	A		600	700		
2.000	B	A		500	500		
2.097152	B	A		500	500		
2.4576	B	A		400	400		
3.000	B	A		150	200		
3.2768	B	A		150	180		
3.579545	B	A		120	180		
3.600	B	A		100	160		
3.6864	B	A		100	140		
4.000	B	A		50	80		
4.096		A			80		
4.194304		A			80		
4.433619		A			80		
4.9152		A			60		
5.000		A			60		
5.0688		A		60			
5.185		A		60			
5.7143		A		50			
5.9904		A		50			
6.000		A		50			
6.144		A		40			
6.400		A		40			
6.5536		A		40			
7.000		A		30			
7.15909		A		30			
7.3728		A		30			
8.000	A		30				
9.216	A		30				
9.8304	A		30				
10.000		A	C		25	50	
10.738635		A	C		25	50	
11.000		A	C		25	50	
11.050		A	C		25	50	

FREQUENCY MHz	HOLDER TYPE			ESR		
	HC-51/U "B"	HC-49/U "A"	UM-1 "C"	"B"	"A"	"C"
11.059		A	C		25	50
12.000		A	C		25	40
12.288		A	C		25	40
13.5168		A	C		25	40
14.31818		A	C		25	35
14.7456		A	C		25	35
15.000		A	C		25	35
16.000		A	C		25	30
16.384		A	C		25	35
17.360		A	C		25	30
18.000		A	C		20	25
18.432		A	C		20	30
19.354		A	C		20	30
19.6608		A	C		20	30
20.000		A	C		20	30
22.1184		A	C		20	25
23.400		A	C		20	25
24.000		A	C		20	25
24.576		A	C		30	25
25.000		A	C		30	20
27.000		A	C		30	20
30.000		A	C		30	20
32.000		A	C		30	60
36.000		A	C		30	60
40.000		A	C		30	50
48.000		A	C		30	50
60.000		A	C		30	40
70.000		A	C		30	40
80.000		A	C		30	40
90.000		A	C		40	40
100.000		A	C		40	100
150.000		A	C		50	70
200.000		A	C		100	120
300.000		A	C		150	180

- **CARRIER TAPE DIMENSIONS**



- **PACKAGING**

370 mm REEL DIAMETER, 18 mm TAPE WIDTH, 12.7 mm PITCH, QUANTITY 1000 PIECES PER REEL

Railtron Electronics/RAMI Technology USA, LLC, including its affiliates, employees, agents and other persons acting on its behalf (collectively Railtron/RAMI Tech), disclaim any and all liability for any errors or inaccuracies contained in this data sheet. While Railtron/RAMI Tech has made every reasonable effort to ensure the accuracy of all product information, specifications and data contained herein, Railtron/RAMI Tech does not guarantee that the information is accurate, reliable or current. The product information is provided for reference purposes only and is subject to change, correction or revision, at any time without notice. Railtron/RAMI Tech does not assume any liability arising out of an application or use of any product described herein and disclaims any warranties expressed or implied. The user of products in such applications shall assume all risks of such use and will agree to hold Railtron/RAMI Tech, harmless against all damages.